One World, Two Protocols

The Quantum–Classical Divide as a Change of Protocol, Shown on a Single Entity Under One Law

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Abstract

Is there a physical divide between quantum and classical physics, or only a change of protocol? We take one entity, a spin of size j, under one Hamiltonian, and read it through different protocols.

One law. For a Hamiltonian linear in the su(2) generators, the expectation values follow the classical rotation law *exactly*, at every j. Only the Lie algebra is used, and the numerical error is below 10^{-10}.

Fine protocol. A sharp parity measurement violates the Leggett–Garg inequality: $K = 1.199$, 1.195, 1.192 and 1.192 for $j = 5$, 10, 20, 40, against the macrorealist bound 1. No classical (macrorealist) description exists for this protocol.

Coarse protocol. The same entity, read by a fuzzy sign of J_z with resolution $2\sqrt{j}$, gives $K = 0.545$, 0.786, 0.950 and 0.997, all at most 1. It is nearly non-invasive: an earlier coarse reading changes a later statistic by 5×10^{-4} and 5×10^{-5} at $j = 10$ and 40, against 0.36 and 0.47 after a parity reading. Ordinary macro law appears.

Where coarse-graining is not enough. Under nonlinear (twisting) dynamics, the coarse mean agrees with a classical ensemble early on. At $t = \pi j$, however, the quantum mean is -1 while the classical one is $+1$. An unobserved environment (dephasing) drives it toward 0. Classicality then needs a second protocol choice, namely ignoring the environment.

Collapse. The nonselective measurement equals a unitary interaction with a pointer followed by a partial trace. "Collapse" is the conditional record of a protocol, and the law on system and pointer stays unitary.

What this shows, and what it does not. Within quantum theory, no physical divide is needed for these entities: one law, with protocols of different resolution and different environment bookkeeping. The result rests on established work (Leggett–Garg; Kofler–Brukner; decoherence). RG's contribution is the typing:

- macro law is the law *descended* to the coarse quotient;
- quantum features are *hidden* values that only fine protocols read;
- decoherence is the environment as an *unobserved* protocol.

It does not settle the measurement problem, it assumes the universality of quantum theory, and it does not test gravity-scale superpositions. It is strong evidence against a *physical* divide. It is not a coffin for every question about one.

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1 The claim, stated so that it can be tested

Definition 1.1 (Protocol divide)**.** A *protocol divide* holds for an entity if there are one state space and one dynamical law, together with two protocol families, fine and coarse, such that:

- the fine family admits no macrorealist description (Leggett–Garg is violated);
- the coarse family is described, within a declared error, by a classical law (Leggett–Garg holds, the readings are non-invasive, and means follow classical equations).

A *physical divide* would instead require different laws at different scales.

Imported Theorem 1.2 (Leggett–Garg [1])**.** *Under macrorealism and non-invasive measurability, a dichotomic quantity $Q = \pm 1$ measured in pairs at times $t_1 < t_2 < t_3$ satisfies $K = C_{12} + C_{23} - C_{13} \leq 1$.*

2 One law

Theorem 2.1 (Exact classical law for Lie-algebraic Hamiltonians)**.** *If $H = \boldsymbol{\omega} \cdot \mathbf{J}$, then $\frac{d}{dt}\langle \mathbf{J} \rangle = \boldsymbol{\omega} \times \langle \mathbf{J} \rangle$ exactly, for every spin j and every state.*

Proof. The Heisenberg equation uses only $[J_a, J_b] = i\varepsilon_{abc}J_c$, so the operator equation is linear in $\mathbf{J}$. Take expectations. Witness (W1): for $j = 7.5$ and $H = 0.8J_x$, the computed $\langle J_z \rangle = j\cos(0.8t)$ and $\langle J_y \rangle = -j\sin(0.8t)$ hold to $< 10^{-10}$ on 25 times. □

The *law* is the same at every j. What differs between "quantum" and "classical" is not the equation of motion. It is what a protocol can read: fluctuations, correlations, and the effect of reading.

3 Fine and coarse protocols on the same entity

Take $H = J_x$, initial state $|j, j\rangle$ (a coherent state along $+z$), and pairwise readings with Lüders updates.

Definition 3.1 (The two protocols)**.**

- *Fine:* parity $Q = (-1)^{j-m}$ of J_z, read sharply.
- *Coarse:* the sign of J_z, read through a fuzzy POVM $E_\pm(m) = \Phi(\pm m/\Delta)$ with resolution Δ.

Proposition 3.2 (Leggett–Garg, executed)**.** *Maximizing over the reading interval $\tau \in (0, \pi]$:*

j	*fine parity* K	*coarse* $\Delta = \sqrt{j}$	*coarse* $\Delta = 2\sqrt{j}$	*coarse* $\Delta = 3\sqrt{j}$
5	*1.199*	*0.953*	*0.545*	*0.297*
10	*1.195*	*0.997*	*0.786*	*0.502*
20	*1.192*	*0.99999*	*0.950*	*0.747*
40	*1.192*	*1.0000000*	*0.997*	*0.931*

The fine protocol violates the inequality by about 0.19 *at every size. The coarse protocol satisfies it once $\Delta \gtrsim 2\sqrt{j}$. At $\Delta = \sqrt{j}$ it sits at the boundary as j grows.*

Proposition 3.3 (Invasiveness)**.** *Take readings at $\tau = \pi/3$ and a later coarse statistic at 2τ, with $\Delta = 2\sqrt{j}$:*

j	*change after an earlier parity reading*	*change after an earlier coarse reading*
10	*0.359*	5.4×10^{-4}
40	*0.467*	5.0×10^{-5}

The fine reading disturbs the later statistic. The coarse one does not, and less so as j grows.

Imported Theorem 3.4 (Classicality from coarse-graining [2])**.** *For spin-j systems under rotation-type Hamiltonians, measurements with resolution much larger than $\sqrt{j}$ yield statistics compatible with a classical (Liouville) evolution of a spin vector, while sharp measurements can violate macrorealism at every j.*

Corollary 3.5 (A protocol divide, not a physical one)**.** *For this entity the divide is a protocol divide: one Hamiltonian, one state space, and two read-outs. "Macro law" is the law descended to the coarse quotient. Quantum features (the parity correlations) are hidden values that only the fine protocol reads. The relative fluctuations of the coarse mean are $\propto 1/\sqrt{j}$.*

4 Where coarse-graining is not enough

Proposition 4.1 (Nonlinear dynamics and the environment)**.** *Take the twisting Hamiltonian $H = J_x^2/j$ with initial state $|j, j\rangle$. In the J_x eigenbasis, $\langle J_z\rangle(t)$ is a sum of terms* $\cos((2\lambda{+}1)t/j)$*. A classical ensemble with the same J_x distribution gives* $\cos(2\lambda t/j)$ *instead.*

	t	$j = 10$	$j = 20$	$j = 40$
quantum $\langle J_z\rangle/j$	$\pi j/4$	*0.0014*	10^{-6}	10^{-12}
classical ensemble	$\pi j/4$	*0.0014*	10^{-6}	10^{-12}
quantum	πj	-1.000	-1.000	-1.000
classical ensemble	πj	$+1.02$	$+1.01$	$+1.006$
dephased ($\Gamma = 0.05$)	πj	-0.46	-0.21	-0.04

Reading.

- Early on, coarse quantum and classical agree.
- At $t = \pi j$ the quantum state is a reversed coherent state, a coarse-grained quantum effect that comes from the discreteness $2\lambda + 1$. No classical ensemble shows it.
- An unobserved environment that monitors J_x erases it, increasingly so as j and t grow.

Classicality of nonlinear dynamics therefore needs two protocol choices: a coarse read-out *and* an untracked environment. Both are protocol choices. The law on system and environment remains unitary. This agrees with Kofler–Brukner's conditions for violations of macrorealism [3] and with decoherence theory [4].

5 Collapse as a protocol record

Theorem 5.1 (Measurement dilation; von Neumann, Ozawa [5, 6])**.** *Every quantum instrument can be realized as a unitary interaction of the system with a pointer, followed by a read-out of the pointer. In particular, the nonselective measurement channel equals a unitary on system and pointer followed by a partial trace.*

Witness (W5). For $\psi = 0.6|0\rangle + 0.8i|1\rangle$, a CNOT onto a pointer in state $|0\rangle$, followed by a trace over the pointer, gives exactly $P_0\rho P_0 + P_1\rho P_1$.
RG typing. "Collapse" is the conditioning of a state on a protocol's record, i.e. the Lüders update given the outcome. Averaged over records it is a completely positive channel with a unitary dilation. It changes what the record supports, not the dynamical law. Which outcome occurs is *not* explained. RG is neutral between interpretations (Everett, Bohm, collapse models, QBism) on that point.

6 What this locks, and what remains

Claim	Status	Basis
One law: classical equations for means under Lie-algebraic H, at every j	THEOREM	Section 2
Fine protocol on the same entity violates macrorealism	EXECUTED	LG $K \approx 1.19$
Coarse protocol on the same entity is classical and non-invasive	EXECUTED + IMPORTED	LG ≤ 1; invasiveness $\to 0$; KB07
Nonlinear dynamics needs environment bookkeeping	EXECUTED	twisting at $t = \pi j$
Collapse is a protocol record with a unitary dilation	THEOREM (imported)	von Neumann, Ozawa
No physical divide is needed for these entities	CONDITIONAL	assumes universal quantum theory
The measurement problem (which outcome)	NOT ADDRESSED	interpretation-neutral
Gravity-scale superpositions	UNTESTED	open experimentally

Experimental status. Leggett–Garg violations have been observed in macroscopic or mesoscopic systems, for example superconducting circuits [7]. Matter-wave interference has been demonstrated with molecules above 25,000 amu [8]. No physical divide has been found. This supports, but does not prove, the protocol reading.

Why this is not a coffin. The paper shows that *no physical divide is needed* wherever quantum theory is universal and protocols are declared. It cannot exclude a new law at scales not yet tested, such as gravitational collapse models. It does not resolve which outcome a record shows. A defensible title is "no divide is *required*", not "no divide *exists*".

7 Reproduction and limits

- `rg_one_world.py` (NumPy, SciPy, SymPy) runs W1–W5. It is deterministic, and reruns are byte-identical.
- *Limits.*

– One family of entities (spin $j \leq 40$), two Hamiltonians, and one choice of fine and coarse observables.
– Leggett–Garg is maximized over a grid of 240 intervals.
– The environment is a single dephasing channel.
– The general classicality and dilation theorems are imported.

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